[Def. A Homotopy of paths in X is a family

$f_t : I \to X$, $0 \le t \le 1$ such that

1) For all $0 \le t \le 1$, $f_t(0) = x_0$ and $f_t(1) = x_1$.
2). $F : I \times I \to X ; (s,t) \mapsto f_t(s)$ is continuous.

Given two path p and q from $x_0 \in X$ to x_1, they are said to be homotopic if: there exises a hopotopy of paths f_t satisfying $p = f_0$ and $q = f_1$.

Denoted as $p \simeq q$.

[Example. Linear Homotopy

Given two path p and q in $\mathbb{R}^n$ having the same endpoint x_0 and x_1, define

$$f_t : I \to \mathbb{R}^n ; s \mapsto (1-t)p(s) + t\,q(s)$$

for each $0 \le t \le 1$. Then, given t,

$$f_t(0) = (1-t)p(0) + t\,q(0) = x_0 \quad \text{and} \quad f_t(1) = (1-t)p(1) + t\,q(1) = x_1$$

(In actually, for any $s \in [0,1]$, $p(s) = q(s) \Rightarrow f_t(s) = p(s)$)

And, to show $F : I \times I \to \mathbb{R}^n ; (s,t) \mapsto f_t(s)$ is continuous, let $M > 0$ be a bound for p and q.

Let $(s_0, t_0) \in I \times I$ be given. Given $\varepsilon > 0$, there is $\delta > 0$ such that

$$\|s - s_0\| < \delta \Rightarrow \|f_{t_0}(s) - f_{t_0}(s_0)\| < \varepsilon \text{ and}$$

$$\begin{aligned} \|t - t_0\| < \delta \Rightarrow \|f_t(s) - f_{t_0}(s)\| &= \|(1-t)p(s) + t\,q(s) - [(1-t_0)p(s) + t_0\,q(s)]\| \\ &= \|(t_0 - t)p(s) + (t - t_0)q(s)\| \\ &\le |t_0 - t| \cdot (\|p(s)\| + \|q(s)\|) < \varepsilon. \end{aligned}$$

Now,

$$\|(s,t) - (s_0,t_0)\| < \delta \Rightarrow \|f_t(s) - f_{t_0}(s_0)\| \le \|f_t(s) - f_{t_0}(s)\| + \|f_{t_0}(s) - f_{t_0}(s_0)\| < 2\varepsilon.$$

So $p \simeq q$ ∎

Prop. Let X be a Topological space. Given two points x_0 and x_1 in X, define an relation on the set of paths from x_0 to x_1 as: $p \sim q \iff p \simeq q$.
Then, this relation is equivalence relation.
Proof. Reflexivity is evident, by $f_t = f$, the constant homotopy.
Symmetry is given by the inverse homotopy: if $f_0 \simeq f_1$ via f_t, then $f_1 \sim f_0$ via f_{1-t}.
To show the Transivity, suppose that $f_0 \simeq f_1$ via f_t and $g_0 \simeq g_1$ via g_t, with $f_1 = g_0$.
Define $H: I \times I \to X : (s,t) \mapsto \begin{cases} f_{2t}(s) & \text{if } 0 \le t \le \frac{1}{2} \\ g_{2t-1}(s) & \text{if } \frac{1}{2} \le t \le 1. \end{cases}$
It is well-defined since $f_1 = g_0$. And, Continuity is given by the Pasting Lemma.

Def. A path $p: I \to X$ is called a loop if $p(0) = p(1) = x_0 \in X$,
and the point x_0 is called the basepoint.
The collection of all homotopy classes $[f]$ of loops $f: I \to X$ at the basepoint x_0,

$$\pi_1(X, x_0) \overset{\text{def}}{=} \{ [f] \mid f \text{ is a loop at } x_0 \}$$

is called the fundamental group under the operation

$$[f][g] \overset{\text{def}}{=} [f \cdot g] \quad \text{where } f \cdot g : I \to X : s \mapsto \begin{cases} f(2s) & \text{if } 0 \le s \le \frac{1}{2} \\ g(2s-1) & \text{if } \frac{1}{2} \le s \le 1 \end{cases}$$

This $\pi_1(X, x_0)$ is well-defined as a set clearly, and the operation is also well defined by:
Suppose that $[f_1] = [f_2]$ and $[g_1] = [g_2]$. Then $f_1 \simeq f_2$ and $g_1 \simeq g_2$ via $\hat{f}_t$ and $\hat{g}_t$.

$$[f_1][g_1] = [f_1 \cdot g_1] = [f_2 \cdot g_2] \text{ because } f_1 g_1 \simeq f_2 g_2 \text{ via } \hat{f}_t \cdot \hat{g}_t$$
$$= [f_2][g_2].$$

Moreover, the operation satisfies the Group Axiom, the proof lies in the next Page.

proof of the fundamental group.

First, given a path $f: I \to X$, define a reparametrization of a path f as a composition

$$f \circ \varphi \quad \text{where} \quad \varphi: I \to I \text{ is any continuous map s.t } \varphi(0)=0 \text{ and } \varphi(1)=1.$$

The reparametrization preserves homotopy, that is,

$f \circ \varphi \simeq f$ via $f \circ \varphi_t$ where $\varphi_t(s) = (1-t)\varphi(s) + ts$, so that $\varphi_0 = \varphi$ and $\varphi_1 = s$.

「Associativity. Let f, g, h be paths with $f(1) = g(0)$ and $g(1) = h(0)$.

Then, the composition $(f \cdot g) \cdot h = f \cdot (g \cdot h)$ are defined. Moreover,

$f \cdot (g \cdot h)$ is a reparametrization of $(f \cdot g) \cdot h$, because

$$f \cdot (g \cdot h)(s) = \begin{cases} f(2s) & \text{if } 0 \le s \le \frac{1}{2} \\ g(4s-2) & \text{if } \frac{1}{2} \le s \le \frac{3}{4} \\ h(4s-3) & \text{if } \frac{3}{4} \le s \le 1 \end{cases} \quad \text{and}$$

$$(f \cdot g) \cdot h(s) = \begin{cases} f(4s) & \text{if } 0 \le s \le \frac{1}{4} \\ g(4s-1) & \text{if } \frac{1}{4} \le s \le \frac{1}{2} \\ h(2s-1) & \text{if } \frac{1}{2} \le s \le 1 \end{cases} \quad \text{therefore}$$

$f \cdot (g \cdot h) = (f \cdot g) \cdot h \circ \varphi$ where $\varphi(s) = \begin{cases} \frac{1}{2}s & \text{if } 0 \le s \le \frac{1}{2} \\ s - \frac{1}{4} & \text{if } \frac{1}{2} \le s \le \frac{3}{4} \\ 2s-1 & \text{if } \frac{3}{4} \le s \le 1 \end{cases}$

so $f \cdot (g \cdot h) \simeq (f \cdot g) \cdot h$, the associative is proven

「Identity. Given a path $f: I \to X$, let C be the constant path, as

$$C_i: I \to X; s \mapsto f(i) \quad (i = 0, 1)$$

Then $f \cdot C_1$ is a reparametrization of f via

$$\varphi: I \to I; s \mapsto \begin{cases} 2s & \text{if } 0 \le s \le \frac{1}{2} \\ 1 & \text{if } \frac{1}{2} \le s \le 1. \end{cases}$$

and similarly $C_0 \cdot f$ is also a reparametrization, so, $C_0 f \simeq f \cdot C_1 \cong f$.

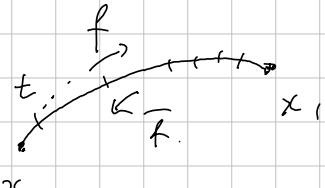
For a loop, the endpoints are fixed, so $C(s) = x_0$ is two-sided identity in $\pi_1(X, x_0)$.

「Inverse. Given a path f from x_0 to x_1, define the inverse path

$$\overline{f}(s) = f(1-s) \quad \text{so that } \overline{f}(0) = f(1), \ \overline{f}(1) = f(0)$$

Then $f \cdot \overline{f}$ is Homotopic to a constant path $C(s) = f(0)$, via $h_t = f_t \cdot \overline{f}_t$ where

$$f_t: I \to X; s \mapsto \begin{cases} f(s) & \text{if } 0 \le s \le 1-t \\ f(1-t) & \text{if } 1-t \le s \le 1 \end{cases}$$



similarly, for a loop, $f \cdot \overline{f} \simeq \overline{f} \cdot f \simeq C (\equiv x_0)$.

Change of basepoint.

Def. Let X be a space, and $x_0, x_1 \in X$ be points contained a same path-component. Let $h: I \to X$ be a path from x_0 to x_1, with the inverse curve $\overline{h}(s) = h(1-s)$. Then, for any loop at x_1, the composition $h \cdot f \cdot \overline{h}$ is a loop at x_0. Rigorously, $(h \cdot f) \cdot \overline{h} \neq h \cdot (f \cdot \overline{h})$ but $(h \cdot f) \cdot \overline{h} \simeq h \cdot (f \cdot \overline{h})$. Therefore, the map

$$\beta_h : \pi_1(X, x_1) \to \pi_1(X, x_0) : [f] \mapsto [h \cdot f \cdot \overline{h}]$$

is well-defined, and is called a change of basepoint.

Moreover, for path-connected x_0 and x_1, the map β_h is an Isomorphism.

Holomorphism: $\beta_h[f \cdot g] = [h \cdot f \cdot g \cdot \overline{h}] = [h \cdot f \cdot \overline{h} \cdot h \cdot g \cdot h] = \beta_h[f]\beta_h[g]$.

Invertible: $\beta_{\overline{h}}$ satisfies

$$\beta_h \beta_{\overline{h}}[f] = \beta_h[\overline{h} \cdot f \cdot h] = [h \cdot \overline{h} \cdot f \cdot h \cdot \overline{h}] = [f].$$

and similarly $\beta_{\overline{h}} \beta_h[f] = [f]$.

Def. A space X is called if it is path-connected and has trivial fundamental group.

Def. Given a space X, a covering space of X consists of a space $\tilde{X}$ and a map $p: \tilde{X} \to X$ s.t. For each $x \in X$, there is an open neighborhood U of x in X satisfying

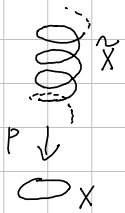
$$p^{-1}[U] = \bigsqcup_{i \in \Lambda} \mathcal{O}_i, \quad \text{each } \mathcal{O}_i \text{ s.t. } \mathcal{O}_i \overset{\text{Homeo}}{\simeq} U \text{ via } p.$$

Example. Given a space S^1, let $p: \mathbb{R} \to S^1 : s \mapsto (\cos 2\pi s, \sin 2\pi s)$. Let $(1,0) \in S^1$ be given. Then, for a neiborhood $B_1(1,0) \cap S_1$

$$p^{-1}[B_1(1,0) \cap S^1] = \bigsqcup_{n \in \mathbb{Z}} \left(n + \left(-\tfrac{1}{8}, \tfrac{1}{8}\right)\right)$$

and for each $n \in \mathbb{Z}$,

$$\left(n + \left(-\tfrac{1}{8}, \tfrac{1}{8}\right)\right) \simeq B_1(1,0) \cap S^1 \text{ via } p.$$



Prop. Let X and Y are Path-Connected. Then,

$$\pi_1(X\times Y) \cong \pi_1(X)\times\pi_1(Y)$$

Proof. Notice that: $f: Z\to X\times Y$ is continuous iff $p_1\circ f$ and $p_2\circ f$ both are continuous, where $p_1: X\times Y\to X$ and $p_2: X\times Y\to Y$ are projections.
Therefore, there is a one-to-one correspondence

$$\{f: I\to X\times Y \mid f \text{ is a loop at } (x_0,y_0)\}$$

$$\Updownarrow$$

$$\{g: I\to X \mid g \text{ is a loop at } x_0\in X\}\times\{h: I\to Y \mid h \text{ is a loop at } y_0\in Y\},$$

Under this correspondence, we can define a bijective

$$\pi_1(X\times Y,(x_0,y_0))\to \pi_1(X,x_0)\times\pi_1(Y,y_0)$$

$$: \quad [f]\mapsto([g],[h]) \quad \text{where } f=(g,h)$$

is a bijective group Homomorphism, so proved

The Fundamental Group of the Circle.

Thm. $\pi_1(S^1)$ is an infinite cyclic group generated by $[w]$, where

$$w(s) = (\cos 2\pi s, \sin 2\pi s) \text{ based at } (1,0).$$

Note that $[w]^n = [w_n]$ where $w_n(s) = (\cos 2\pi ns, \sin 2\pi ns)$ for each $n \in \mathbb{N}$.
Consider $p: \mathbb{R} \to S^1; s \mapsto (\cos 2\pi s, \sin 2\pi s)$ and $\tilde{w}_n: I \to \mathbb{R}; s \mapsto ns$.
Then p is a covering map, and $\tilde{w}_n$ is called the lift of w_n satisfying

$$w_n = p \circ \tilde{w}_n.$$

Proof. Let $f: I \to S^1$ be a loop at the basepoint $x_0 = (1,0)$.
Since S^1 is path-connected, $[f] \in \pi_1(S^1)$.

[Thm. Let X be a space and let $w: I \to X$ be a path from x_0 to x_1. Then, a map

$$\hat{w}: \pi_1(X, x_0) \to \pi_1(X, x_1)$$
$$: [\alpha] \mapsto [\bar{w} * \alpha * w]$$

is a Group Isomorphism.

Thm. Let $f: X \to Y$ be a continuous map such that $f(x_0) = y_0$.
Then, f induces a group Homophism

$$f_\# : \pi_1(X, x_0) \to \pi_1(Y, y_0) ; [\alpha] \mapsto [f \circ \alpha].$$

Prof. Notice that given a loop $\alpha : I \to X$ at x_0, a composition $f \circ \alpha : I \to Y$ satisfies $(f \circ \alpha)(0) = (f \circ \alpha)(1) = f(x_0) = y_0$, so $f \circ \alpha$ is a loop at y_0.
Let $[\alpha], [\beta] \in \pi_1(X, x_0)$. If $[\alpha] = [\beta]$, then there is a Homotopy
$F : I \times I \to X$ s.t $F(x,0) = \alpha(x)$ and $F(1,0) = \beta(x)$, so that

$f \circ F : I \times I \to Y$ satisfies $(f \circ F)(x,0) = (f \circ \alpha)(x)$ and $(f \circ F)(x,1) = (f \circ \beta)(x)$
therefore, $\alpha \overset{F}{\simeq} \beta$ implies $f \circ \alpha \overset{f \circ F}{\simeq} f \circ \beta$, thus $[f \circ \alpha] = [f \circ \beta]$.
It proves the $f_\#$ is well-defined. Now,

$$f_\#([\alpha] * [\beta]) = f_\#([\alpha * \beta]) = [f \circ (\alpha * \beta)] = [(f \circ \alpha) * (f \circ \beta)] = [f \circ \alpha] * [f \circ \beta]$$
$$= f_\#([\alpha]) * f_\#([\beta])$$ ∎

Thm. Let $f : X \to Y$ be a Homotopy equivalence such that $f(x_0) = y_0$.
Then, the induced map $f_\#$ is an isomorphism.